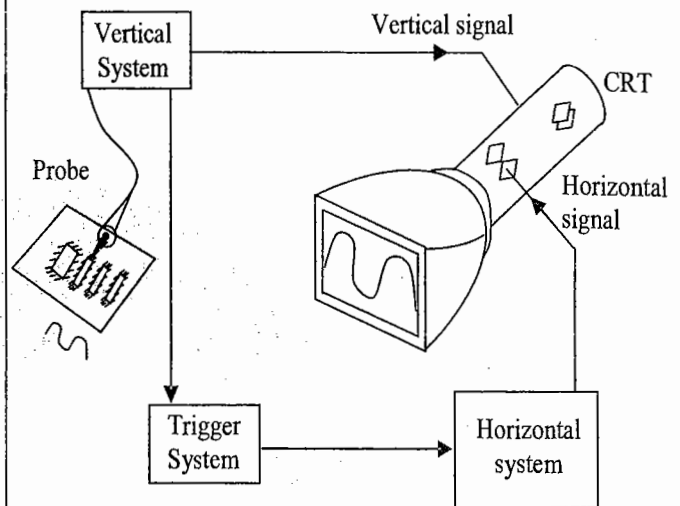


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Cathode Ray Oscilloscope

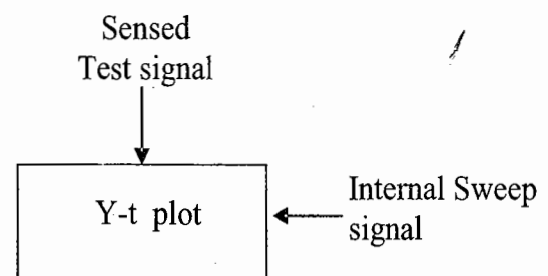
- The cathode ray oscilloscope is an electronic instrument that presents a high-fidelity graphical display of the rapidly changing voltage at its input terminals. The cathode ray oscilloscope is probably the most versatile and useful instrument available for signal measurement.
- Unlike meters, which only allow the user to measure amplitude information, the oscilloscope allows the user to view the instantaneous voltage versus time such displayed plot of the signal can be used for various measurements like peak voltage, frequency, phase, time period, rise time etc. It can also indicate the nature and magnitude of noise that may be corrupting the measurement signal. The more expensive models can measure signals at frequencies up to 500MHz and even the cheapest models can measure signals up to 20 MHz. One particularly strong merit of the oscilloscopes is its high input impedance, typically $1M\Omega$, which means that the instrument has a negligible loading effect in most measurement situations. As a test instrument, it is often required to measure voltages whose frequency and magnitude are totally unknown.
- However, it is not a particularly accurate instrument and is best used where only an approximate measurement is required. Further disadvantages of oscilloscopes include their fragility (being built around a cathode ray tube) and their moderately high cost.
- There are two main classes of oscilloscopes: analog oscilloscopes and digital oscilloscopes.

- A simplified block diagram of an analog oscilloscope is shown in below figure



- The display section of the oscilloscope (i.e., cathode ray tube) has two inputs to it namely vertical(Y) input and Horizontal(X) input. The signals applied to these inputs are driven to corresponding deflection plates and control the position of the electron beam that plots the waveform on the screen. There are two types of plots that can be displayed based on mode of operation of oscilloscope.

1) Sweep mode of operation:

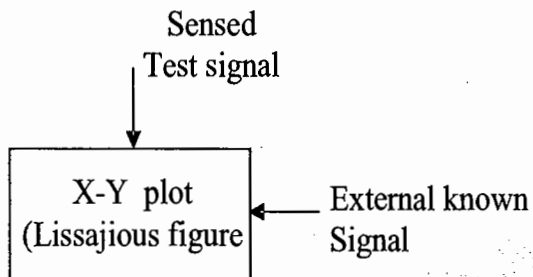


Using
Y-t plot



Various measurements regarding the test signal can be carried out like peak voltage, frequency, phase, time period, resistive etc.

2) X-Y Display mode of operation:



Using Lissajous figure, two measurements can be carried out namely frequency measurement and phase measurement.

• Other important measurements that can be made by oscilloscope are:

- i) Measurement of impedance
- ii) Measurement of power and power factor
- iii) Measurement of degree of modulation
- iv) Measurement of checking of switching times
- v) Tracing of hysteresis loop for magnetic material
- vi) Study of transistors phenomenon
- vii) Fault testing of windings of electrical machines
- viii) Determination of characteristics of thermionic values

Oscilloscope Specifications:

- 1) **Sensitivity:** a series of attenuators and pre-amplifiers exist at the input to the oscilloscope. These condition the measured signal to the optimum magnitude for input to the main amplifier and vertical deflection plates, thus enabling the instrument to measure a very wide range of different signal magnitudes. Selection of the appropriate input amplifier/attenuator is made by setting a

VOLT/DIV control associated with each oscilloscope channel. This defines the magnitude of the input signal that will cause a deflection of one division on the screen.

- 2) **Bandwidth:** One important oscilloscope specification is related to the speed of the waveforms that can be measured. This is determined by the bandwidth of the oscilloscope and it is found that the capability of the oscilloscope to accurately display the waveform falls off with increasing frequency. The Bandwidth is defined as the range of frequencies over which the oscilloscope amplifier gain is within 3dB of its peak value. The oscilloscope specification for bandwidth will typically be quoted in the format: Bandwidth = -3dB at 1500 MHz. The -3 dB specification means that an oscilloscope with a specified inaccuracy of $\pm 2\%$ and bandwidth of 100MHz will have an inaccuracy of $\pm 5\%$ when measuring 30MHz signals, and this inaccuracy will increase still further at higher frequencies. Thus, when applied to signal-amplitude measurement, the oscilloscope is only usable at frequencies up to about 0.3 times its specified bandwidth. In most oscilloscopes, the amplifier is direct coupled, which means that it amplifies d.c. voltages by the same factor as low-frequency a.c. ones. For such instruments, the minimum frequency measurable is zero and the bandwidth can be interpreted as the maximum frequency where the sensitivity (deflection/volt) is within 3 dB of the peak value. In all measurement situations, the oscilloscope chosen for use must be such that the maximum frequency to be measured is well within the bandwidth.

- 3) **Risetime:** The rise time is the transit time between the 10% and 90% levels of the response when a step input is applied to the oscilloscope. The oscilloscope must have a



sufficiently fast rise time to capture the rapid transitions accurately, otherwise important information may not be displayed and the results could be misleading. Oscilloscopes are normally designed such that: $\text{Bandwidth} \times \text{Rise time} = 0.35$

Thus, for a bandwidth of 100 MHz, rise time $= 0.35/100\,000\,000 \Rightarrow 3.5\text{ns}$.

- Types of Oscilloscopes:

- 1) **Analog Oscilloscope:** The analogue oscilloscope is the traditional form of oscilloscope that has been used in laboratories for many years. It relies on analogue techniques and takes in the vertical and sometimes horizontal signals, amplifying them in an analogue format and displaying them on a cathode ray tube.
- 2) **Digital Storage Oscilloscope (DSO):** The digital storage oscilloscope (DSO) is the conventional form of digital oscilloscope. It uses a raster type screen like that used on a computer monitor or television and in this way displays an image that fills the screen and may include other elements in addition to the waveform. These additional items may include text on the screen and the like.
- 3) **Digital Phosphor Oscilloscope (DPO):** The digital phosphor oscilloscope (DPO) is a highly versatile form of oscilloscope that uses

a parallel processing architecture to enable it to capture and display signals under circumstances that may not be possible using a standard DSO. The key element of a DPO is that it uses a dedicated processor to acquire waveform images. In this way it is possible to capture transient events that occur in digital systems more easily. These may include spurious pulses, glitches and transition errors. It also emulates the display attributes of an analogue oscilloscope, displaying the signal in three dimensions: time, amplitude and the distribution of amplitude over time, all in real time.

- 4) **Sampling Oscilloscope:** These oscilloscopes are used for analyzing very high frequency signals. They are used for looking at repetitive signals which are higher than the sample rate of the scope. They collect the samples by assembling samples from several successive waveforms, and by assembling them during the processing, they are able to build up a picture of the waveform. The oscilloscope specifications for these items may detail a frequency capability or bandwidth sometimes as high as 50 GHz. However these scopes are very expensive.